FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065
(INTELLIGENT SYSTEMS)

TECHNICAL ASSESSMENT

2

RULE – BASED AGENT

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I. BACKGROUND INFORMATION

A Rule-Based Agent is a type of intelligent agent that operates by following predefined rules to decide its actions. In this lab, students will work with a vacuum cleaner agent that can clean rooms or move between rooms depending on their state (dirty or clean).

The goal is to understand how agents perceive the environment, act according to rules, and update the environment.

II. TECHNICAL ASSESSMENT

RULE – BASED AGENT

Objective

At the end of this laboratory activity, students are expected to:

1. Explain the concept of rule-based agents.
2. Implement a rule-based agent in Python.
3. Interact with an environment by perceiving and acting based on rules.
4. Observe and analyze the behavior of an agent in a controlled environment.

Materials Needed

- Computer with Python 3.x installed
- Text editor or IDE (e.g., VSCode, PyCharm)
- Optional: Jupyter Notebook for interactive testing

General Instructions

1. This **Technical Assessment (TA4)** must be completed individually.
2. Follow the Task below.
3. Include clear comments in your code to describe its logic.
4. Each program should display the correct input, process, and output.
5. Ensure that your script displays clear and readable outputs.
6. Compile all Python codes and corresponding outputs into one PDF file.
7. Save it as **TA2_lastname.pdf**
   (Example: TA2_Patilano.pdf)
8. Submit the **python file** and **PDF** file.

Laboratory Task

Pre-Lab Preparation

1. Review the Python concepts:
   - Classes and objects

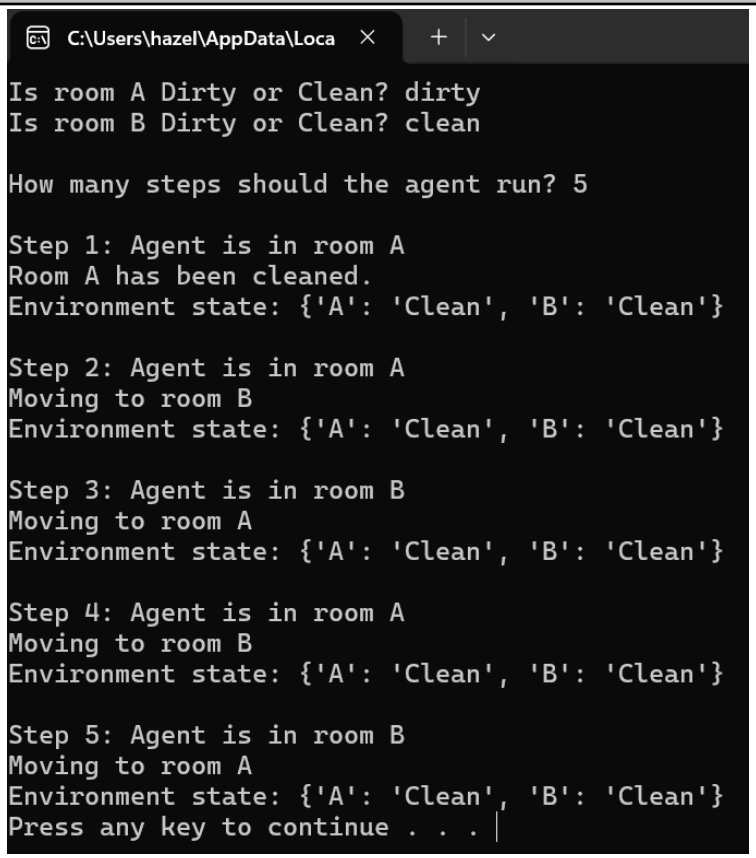
- If-Else statements
- Loops (for loop)
- Functions / methods

2. Understand the environment:
   - Two rooms: A and B
   - Each room can be Dirty or Clean

Task 1: Set Up the Environment

1. Initialize two rooms using a Python dictionary.
2. Allow the user to set the **initial state** of each room (dirty or clean).
3. Write a function to **check if a room is dirty** and a function to **clean a room**.

Sample Output



Task 2: Implement the Rule-Based Agent

1. Create a class for the agent.
2. Implement the **perceive-and-act method**:
   - Rule 1: If the current room is dirty, clean it.
   - Rule 2: If the current room is clean, move to the other room.
3. Implement a **move method** to switch rooms.

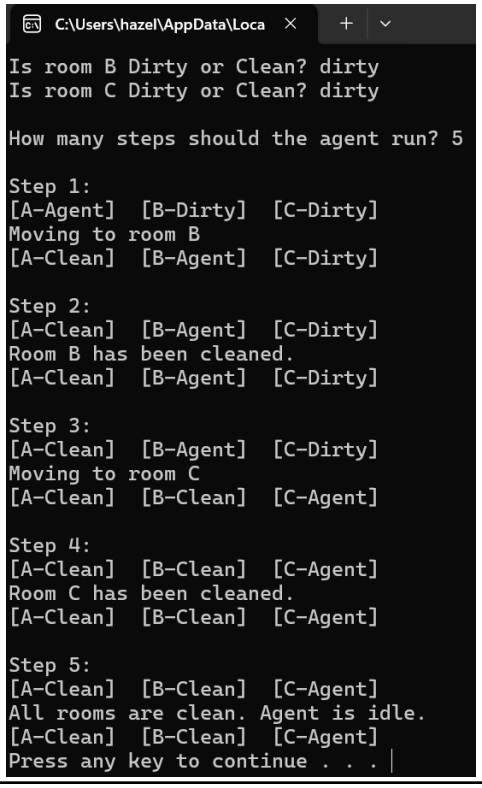
Task 3: Run the Simulation

1. Ask the user how many steps the simulation should run.
2. Run the agent in a loop for the specified steps.
3. After each step, print:
    - The agent's current location
    - The action taken (cleaned/moved)
    - Current state of the environment

Optional (Bonus Task)

- Add a **third room (C)** and update the agent logic.
- Make the agent **randomly choose which dirty room to clean next**.
- Visualize the agent moving between rooms using a **text-based grid**.

Sample Output (Bonus Task)



Your Code - Laboratory Tasks ⇩

```python
import os

# ========================================
# Task 2: Implement the Rule-Based Agent
# ========================================
class ReflexVacuumAgent:
    def __init__(self, initial_location='A'):
        self.location = initial_location

    def move(self):
        """Switches the agent's location between room A and room B."""
        self.location = 'B' if self.location == 'A' else 'A'
        return f"Moving to room {self.location}"

    def perceive_and_act(self, environment):
            """Perceives the current room's state and applies rule-based
actions."""
        current_state = environment[self.location]

        # Rule 1: If the current room is dirty, clean it.
        if current_state == 'Dirty':
            environment[self.location] = 'Clean'
            action_taken = f"Room {self.location} has been cleaned."

        # Rule 2: If the current room is clean, move to the other room.
        else:
            action_taken = self.move()

        return action_taken


# ========================================
# Task 1: Environment Setup & User Inputs
# ========================================
state_a = input("Is room A Dirty or Clean? ").strip().capitalize()
state_b = input("Is room B Dirty or Clean? ").strip().capitalize()
```

```
environment = {
    'A': state_a,
    'B': state_b
}

# ========================================
# Task 3: Run the Simulation
# ========================================
print()
steps = int(input("How many steps should the agent run? "))
print()

# Instantiate the agent starting at Room A
agent = ReflexVacuumAgent(initial_location='A')

for step in range(1, steps + 1):
    # Print agent's location at the start of the step
    print(f"Step {step}: Agent is in room {agent.location}")

    # Agent perceives and chooses action based on Rule 1 or Rule 2
    action = agent.perceive_and_act(environment)
    print(action)

    # Print the current state of the environment
    print(f"Environment state: {environment}")
    print()

os.system("pause")
```

Task 4: Analysis Questions

After running the simulation, answer the following:

1. What rules does the agent follow?
   The agent only moves between the two rooms when the current room they are in is “Clean”. If the room is still dirty, it’ll remain there until it is cleaned.
2. How does the agent decide when to clean and when to move?
   With the use of rule-based actions, when the agent perceives a room to be “Dirty”, it is tasked to clean the room and move to another room to also determine if it is clean or not.
3. What happens if both rooms are already clean at the start?
   The agent moves back and forth between the two rooms.
4. How would you modify the agent to handle **more than two rooms**?
   I have to remove the hardcoded binary check of B and A. I’d put all the rooms in a list and make the agent cycle from A->D for example and wrap back to A until the remaining steps are subsided.

Answer the Questions ⇩